

ONKAR KOTULE

Software Development Engineer II | Full-Stack | Agentic AI | Node.js | React | Python

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PROFESSIONAL SUMMARY

Software Development Engineer with **4+ years of experience** building production-grade backend and full-stack applications across enterprise and startup environments. Strong expertise in **Node.js, TypeScript/JavaScript, React.js, Python, REST APIs, MongoDB, PostgreSQL, Azure, and distributed application design**.

Hands-on experience building **LLM-powered applications and agentic AI workflows** using LangGraph, prompt engineering, structured outputs, response validation, and multi-step AI pipelines. Experienced in designing reliable backend systems with **transaction safety, concurrency handling, real-time communication, workflow orchestration, asynchronous processing, and API-driven architectures**.

Proven ability to own features end-to-end—from architecture and development through CI/CD, production deployment, monitoring, and support—with a strong focus on **scalability, reliability, observability, and maintainable system design**.

TECHNICAL SKILLS

Languages: JavaScript (ES6+), TypeScript, Python, Java

Frontend: React.js, Next.js, React Native, HTML5, CSS3, Redux Toolkit, TanStack Query, Tailwind CSS

Backend & APIs: Node.js, Express.js, REST APIs, GraphQL, MongoDB, PostgreSQL, JWT, OAuth, API Integration, , Socket.IO, Transaction Safety, Event-Driven Architecture, etc

AI / Agentic AI: LLM APIs, LangGraph, Prompt Engineering, Agentic Workflows, Multi-Step AI Pipelines, Structured Outputs, Response Validation, AI Tool Integration, AI Evaluation Concepts

Cloud & DevOps: Microsoft Azure, Azure DevOps, CI/CD, Docker, Git, GitHub, Firebase, Appwrite

Testing & Engineering: Jest, Unit Testing, Error Handling, Logging, Production Monitoring, Agile / SDLC

PROFESSIONAL EXPERIENCE

Associate Consultant — Software Engineer, Backend & Full-Stack

Capgemini — Pune, India | Jul 2023 – Present

- Designed and developed **enterprise-scale full-stack applications** across backend services and frontend experiences, owning features from requirements and architecture through production deployment and support.
- Built scalable **Node.js backend services and REST API layers** for communication between application components and enterprise systems.
- Developed responsive frontend experiences using **React.js**, optimizing state management, API-driven data flow, component reuse, and UI performance.
- Designed secure application architectures incorporating **role-based access control (RBAC), authentication, auditability, and controlled data flows**.
- Worked on AI-powered applications using **LLM APIs, prompt engineering, structured outputs, response validation, and multi-step AI workflows**.

Key Projects & Contributions

AI-Powered Investment Insights System

- Designed an **LLM-powered pipeline** for generating structured investment insights and reports from application data.
- Implemented **prompt engineering and response validation** to improve consistency, reliability, and controllability of LLM-generated outputs.

- Built multi-step AI processing workflows using **LangGraph**, establishing an agentic workflow pattern for sequential processing and decision-oriented tasks.
- Implemented structured AI responses that could be consumed reliably by backend services and frontend components.
- Developed dynamic frontend components to render **AI-generated insights and intermediate application data** through API-driven interfaces.
- Integrated AI-generated insights into enterprise applications for **real-time user consumption**.
- Applied practical concepts including **context management, structured outputs, error handling, fallback strategies, and deterministic validation** around LLM responses.

ShareSpace — Document Management System

- Designed a hierarchical document management system with optimized data modeling for scalable file and folder organization.
- Built secure upload/download workflows with **access control, lifecycle management, soft delete, and restore capabilities**.
- Designed backend APIs and data flows with emphasis on reliability, security, and maintainability.

Dangerous Goods Transport System

- Designed and developed a **slot-booking and workflow-driven application** supporting operational approval processes.
- Implemented **real-time notifications and approval pipelines** to support asynchronous business workflows.
- Worked with event-driven application patterns to communicate state changes across system components.

Export Control System

- Refactored rigid business logic into a **configuration-driven architecture**, improving maintainability and reducing manual implementation effort.
- Redesigned application flows to make business rules easier to extend and maintain.

Platform & Engineering Contributions

- Implemented **Azure AD SSO** integration for enterprise authentication.
- Built and maintained **CI/CD pipelines using Azure DevOps**.
- Supported production applications as an engineering SPOC, handling **production monitoring, troubleshooting, root-cause analysis, and issue resolution**.
- Collaborated with cross-functional teams on requirements, technical design, development, code reviews, testing, and production releases.

Software Developer — Backend & Full-Stack

BITCS | DigitalFlow | KnockOnce | Jun 2021 – Oct 2022

- Developed scalable backend services using **Node.js, Express.js, and MongoDB**, implementing modular REST APIs with validation, error handling, and reusable service components.
- Built full-stack applications using **React.js, Next.js, Redux Toolkit, and Node.js**.
- Designed optimized frontend data-fetching and caching strategies using **TanStack Query**.
- Improved application performance through optimized state management, component reuse, reduced unnecessary rendering, and efficient API communication.
- Optimized MongoDB performance through **indexing, query optimization, schema design, and aggregation pipelines**.

- Developed production systems with emphasis on **scalability, concurrency, data integrity, and maintainability**.

Expense Management System

- Designed and developed a production-grade expense management platform supporting **multi-user financial transactions**.
- Implemented **transaction-safe balance updates** to maintain consistency across concurrent user operations.
- Built real-time expense tracking and settlement workflows.
- Designed optimized MongoDB data models and aggregation pipelines for reporting and analytics.
- Implemented compound indexes to improve performance for high-frequency queries.
- Designed the system with focus on **concurrency handling, data consistency, transaction safety, and scalable backend architecture**.

Additional Engineering Work

- Built **real-time communication features using Socket.IO and event-driven architecture**.
- Developed modular frontend systems using React/Next.js with optimized state management and data fetching.
- Designed reusable backend components and API integrations for scalable application development.

CERTIFICATIONS

- **Microsoft Certified: Azure Developer Associate (AZ-204)**
- **Microsoft Certified: Azure AI Engineer Associate (AI-104)**
- **Microsoft GenAI Practitioner (AI-900)**

EDUCATION

Bachelor of Engineering (B.E.)

Bharati Vidyapeeth College of Engineering, Pune